Coding & Solution

Date	2 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mann-jain143/Smart-Agricultural
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• Crop recommendation using Machine Learning. • Data preprocessing and analysis. • User-friendly Flask web application. • Real-time crop prediction based on N, P, K, temperature, humidity, pH, and rainfall. • Model evaluation and prediction using trained ML models.
Pending / Incomplete Features	• Cloud deployment of the application. • User authentication and login system. • Fertilizer recommendation module. • Weather API integration for real-time climate data.
Setup / Run Instructions	1. Install Python 3.10+ and required libraries. 2. Clone the GitHub repository. 3. Install dependencies using `pip install -r requirements.txt`. 4. Run the application using `python app.py`. 5. Open `http://127.0.0.1:5000` in a web browser to access the application.

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **OptiCrop** project successfully implements a machine learning-based crop recommendation system using Flask and Scikit-learn. The application analyzes soil nutrients (Nitrogen, Phosphorous, Potassium) and environmental parameters (temperature, humidity, pH, and rainfall) to recommend the most suitable crop. The code follows modular programming practices, making it readable, maintainable, and reusable. Future enhancements include cloud deployment, user authentication, fertilizer recommendation, and integration with real-time weather APIs for improved decision support.